

Bill of Materials

VBF-T4R2-BOM-01 | case t4_r2

Finalist E3-porheadroom | dual caliber g/cell (per cell) and kg/kWh (per 0.0181647 kWh) | 2026-08-26

Binder/conductive-additive rows use the literature-default 96/2/2 wt% split of active mass - annotated, because the Chen2020 parameter set has no binder/aditive keys. Enclosure/tabs/mandrel: Not modeled (no keys).

#	Component	Material / detail	g/cell	kg/kWh	Basis / source
1	Positive active material	NMC811-class, 3.5 um r, por 0.28	0.02	0.001	calc-energy layer_kg_m2 x 0.1027 m2 (r5_E3_energy_dfn.json)
2	Positive conductive additive	carbon, 2 wt% of active (literature default, annotated)	0.00	0.000	96/2/2 split: parameter set has no additive keys
3	Positive binder	PVDF-class, 2 wt% of active (literature default, annotated)	0.00	0.000	96/2/2 split: parameter set has no binder keys
4	Negative active material	graphite-class, 3.0 um r, por 0.28	0.01	0.001	calc-energy layer_kg_m2 x 0.1027 m2 (r5_E3_energy_dfn.json)
5	Negative conductive additive	carbon, 2 wt% of active (literature default, annotated)	0.00	0.000	96/2/2 split
6	Negative binder	CMC/SBR-class, 2 wt% of active (literature default, annotated)	0.00	0.000	96/2/2 split
7	Separator	8 um, 397 kg/m3, por 0.47	0.00	0.000	calc-energy layer_kg_m2 x 0.1027 m2
8	Electrolyte	sigma 1.6 S/m, D 3.2e-10 m2/s, t+ 0.6 (estimates); fill 1.0 x pore volume	0.01	0.000	pore volume 5.0101e-6 m3 x 1.2 g/cm3 (literature default)
9	Positive current collector	Al, 12 um (2700 kg/m3)	0.00	0.000	calc-energy layer_kg_m2 x 0.1027 m2
10	Negative current collector	Cu, 8 um (8960 kg/m3)	0.01	0.000	calc-energy layer_kg_m2 x 0.1027 m2
11	Enclosure / tabs / terminals / mandrel	NOT MODELED	Not modeled	Not modeled	parameter set contains no enclosure/tab keys - honest omission
12	TOTAL cell mass	incl. electrolyte	0.05	0.003	kg/kWh = g/1000 / 0.0181647 kWh (18.1647 Wh, r5_E3_energy_dfn.json)
13	TOTAL without electrolyte	contract caliber per calc-energy mass_kg	0.04	0.002	matches calc-energy mass_kg 0.0395362 (electrolyte excluded)